WEEK 3 · UNIT I · CLO 1 AND CLO 5 · STUDENT PACK

Data Literacy, BI Fundamentals and the Ethics of Adoption

How to read an HR number without being fooled, what a warehouse, metric and dashboard really are, and what collecting data costs the people it describes.

L1 Data literacy: populations and samples, rates and counts, base rates, correlation and cause

L2 BI fundamentals: warehouse, metric, dashboard, and the single source of truth

L3 Ethics and culture of adoption: consent, surveillance, the divide, DPDP and imported tools

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SESSION MAP

Three parts, one argument

60 min

0:00 — 0:22

Data literacy

Population and sample. Rate and count. Correlation and cause. The three places an HR number most often lies.

0:22 — 0:40

BI fundamentals

What a warehouse, a metric and a dashboard actually are — and why a definition is a governance object, not a formula.

0:40 — 1:00

Ethics of adoption

Consent that is not freely given. The surveillance a dashboard makes possible. The divide, seen from the data side.

The argument that joins them: the machinery that lets you answer a question about people is the same machinery that lets you watch them. You cannot build the first and decline the second — you can only govern it.

DATA LITERACY, ONE OF THREE

Who is actually in this number?

Population

Everyone you mean to describe. In HR this is rarely what the file contains — leavers drop out, contractors are excluded, new joiners have not been loaded yet.

Sample

The subset you actually have. A sample supports a claim about the population only if you know *how* it was selected.

Self-selection

The dangerous case. Survey respondents chose to answer; app users chose to log in. The people missing are missing *for a reason connected to the thing you are measuring.*

THE CASE THAT MATTERS HERE

480 of 1,000

The GreenLeaf telemetry covers fewer than half the employee records, and it covers the half that **uses the systems**. Any adoption finding drawn from it describes adopters. The non-adopters you are trying to find are the ones the file cannot see.

DATA LITERACY, TWO OF THREE

A count tells you size. A rate tells you risk.

Claim	What it is	Where it misleads
“Store Operations lost the most people”	A count	The largest department almost always loses the most people. The claim may carry no information at all.
“Store Operations has the highest attrition rate”	A rate — leavers over a base	Only as good as the base. Which headcount, measured when, including whom?
“Attrition rose from 31% to 38%”	A rate over time	Check both parts. A falling denominator raises a rate without a single extra person leaving.

Small denominators

One departure from a team of six is a 17% attrition rate. Ranking departments by rate puts small teams at the top and keeps them there.

Report both, always

A rate without its numerator and denominator cannot be checked, and a rate nobody can check is an assertion in decimal form.

DATA LITERACY, THREE OF THREE

Four things a correlation can be

Two HR variables move together. Before you write a recommendation, rule out the three explanations that are not the one you want.

A causes B

The claim you want. It needs more than co-movement: a mechanism, and ideally variation you did not choose.

B causes A

Reverse causation. Training completion correlates with engagement — and engaged people volunteer for training.

C causes both

A confounder. Grade band drives device access *and* literacy *and* app usage. Three correlations, one cause.

Selection made it

The correlation lives in how the file was built, not in the world. See the previous slide.

Why this is not academic fussiness: an intervention is a causal bet. If you spend on training because training correlates with engagement, and the true story is that engaged people seek out training, you will spend the budget and watch nothing move.

L1 · DATA LITERACY

Base rates, and three HR data traps

50%

The base rate trap

A flight-risk flag is 90% accurate. If 10 of 100 people leave, it flags 9 of the 10 leavers, and also 9 of the 90 stayers. So half of everyone flagged was never going to leave.

Survivorship

Exit interviews hear only from leavers; surveys hear only from people who answer. The silent group is often the one you most need.

Simpson's paradox

A gap can reverse when you split the data. A firm-wide pattern may disappear, or flip, inside every department.

Regression to the mean

The worst store this quarter usually improves next quarter anyway. Do not credit the training you happened to run there.

THE HABIT

Before you believe a percentage, ask: out of how many, compared with what, and who is missing from the count?

BI FUNDAMENTALS

A warehouse is a copy, kept on purpose

THE SOURCES

Operational systems

Payroll, the HRIS, the ATS, the learning platform. Each built to *run* a process, and optimized for today's transaction rather than for last year's history.

THE PIPELINE

Extract, conform, load

Pull the data out, force it onto shared definitions and shared keys, write it down. **The conforming step is where the politics lives.**

THE WAREHOUSE

History that does not move

A separate store, organized for asking questions rather than for running payroll, that remembers what was true in March even after March has been overwritten.

The thing nobody tells you

Operational systems mostly hold the **present**. When a person changes grade, the field is overwritten and the old value is gone. A warehouse exists largely to keep the past — and without it, no trend, no cohort, no predictive model is possible.

L2 · BI FUNDAMENTALS

A single source of truth, and why GreenLeaf has none

What it means

Every report draws on one agreed record for each employee. Ask two managers for headcount and you get the same number.

Two core systems

LegacyHR (on premises, adoption 0.43) and PeopleCloud (pilot, adoption 0.38) both hold employee records.

The spreadsheet paradox

Spreadsheet Trackers: adoption 1.00, satisfaction 1.5 out of 5 (the lowest), zero integrations.

READ THE PARADOX

Universal use with the lowest satisfaction is not success. It is the fallback people use because no system is trusted. Source: TechStack_Inventory.

BI FUNDAMENTALS

A metric is an agreement, not a formula

“Attrition rate” is not a definition. It becomes one only when six further questions have been answered and written down.

Numerator
All leavers, or voluntary only? Retirements? End of fixed term?

Denominator
Opening headcount, closing, or average? Contractors in or out?

Period
Financial year or rolling twelve months? Annualized from a quarter?

Grain
Whole firm, department, site, manager? At what size do you suppress?

Owner
Who may change the definition, and who must be told when it changes?

Restatement
When it changes, is history recalculated or left as published?

Two teams using different answers to any one of these produce different numbers, both correct. **That is not a data-quality problem. It is an absent agreement** — and it cannot be fixed by buying software.

L2 · EXERCISE

Sketch the data you need to measure time to hire honestly

Start the clock

When the vacancy opens, when it is approved, or when it is advertised? Each choice gives a different number.

Stop the clock

Offer made, offer accepted, or first day at work? GreenLeaf's ATS stops at the first joiner's start date.

Choose the summary

Mean or median? Across 140 requisitions the mean is 43.8 days and the median 43. One outlier can move a mean.

Split it

By role, by Location_Tier, by Source_Channel. A single average can hide a slow Tier 3 store channel.

Agree it

Write the definition down and give it an owner. A metric is an agreement, not a formula.

TRY IT

Recruitment_ATS, column Time_to_Fill_Days. Then ask why the Board briefing quotes 47 days for frontline roles.

BI FUNDAMENTALS

A dashboard is a decision surface

Not a report, and not a place to display everything you happen to have. The test of a dashboard is whether someone changes what they do after looking at it.

Every tile needs an owner and an action

If nobody is accountable for the number and nothing follows from it moving, the tile is decoration that costs attention.

Comparison beats level

A bare number cannot be interpreted. Against last period, against a target, against a peer unit — the comparison is what carries the meaning.

A number that is watched gets managed

Publish time-to-fill and time-to-fill improves — sometimes by better hiring, sometimes by opening requisitions later. Measurement changes behavior.

Goodhart's observation

When a measure becomes a target, it stops being a good measure. **Choosing what to put on a dashboard is choosing what the organization will optimize** — including in ways you did not intend.

THE TURN OF THE HOUR

The warehouse that answers your question is the same warehouse that watches your people.

You cannot build one without the other. Individual-level records, joined across systems and kept over time, are the precondition for people analytics *and* the definition of a surveillance capability. The only question ever available is how it is governed.

ETHICS OF ADOPTION

Consent at work is not consent in a shop

Valid consent is informed, specific, and **freely given**. The employment relationship weakens all three, and the third most of all.

Power asymmetry

A person who can be appraised, promoted or released by the party asking is not in a position to refuse comfortably.

Bundling

Consent buried in an onboarding pack, alongside things that genuinely are conditions of employment, is not specific to anything.

Purpose drift

Data given for one purpose becomes a feature in a model built two years later. Consent was for the first use, never the second.

No real exit

If declining means losing access to payslips, leave requests or the roster, the choice is nominal.

What to do instead of leaning on consent

Narrow the collection, state the purpose in words a frontline employee would use, set a retention period and honor it, and give a real route to query or correct. **Design so that consent is not doing work it cannot bear.**

L3 · DPDP ACT 2023 · ORIENTATION

What the law asks of an employer handling staff data

Principle	What it means at work
Notice and consent	Tell people what is collected and why. Consent must be free, specific, informed and clear.
Employment as a legitimate use	Section 7 lets an employer process data for employment purposes without fresh consent. It is not a license for any use.
Purpose limitation	Data collected for payroll should not quietly become data for monitoring or ranking.
Security and breach duty	Reasonable safeguards, and a duty to report a breach to the Data Protection Board and to each person affected.
Rights of the employee	Access, correction and erasure, and a way to raise a grievance.

THE LINK TO ROUND 4 An algorithm trained on employee data inherits every one of these duties.

ETHICS OF ADOPTION

Where measurement becomes monitoring

Rung	Example in an HR stack	What changes at this step
Aggregate, historical	Department attrition by quarter	Nobody is identifiable. Uncontroversial.
Individual, historical	One person's training record and appraisal history	Normal HR administration, and long accepted.
Individual, behavioral	Login times, session length, message volume	The record is now of conduct rather than of outcomes. Presence becomes a proxy for effort.
Individual, inferential	A flight-risk score attached to a named employee	A prediction about the future becomes part of the present record — and can act on the person before the event.
Continuous, ambient	Location, keystrokes, camera, sentiment	Observation without a stated occasion. The employee cannot know when they are not being read.

No single step feels like a decision. **Each is a small extension of the one before, and the step from rung three to rung four is usually taken by an analyst, not by a board.**

ETHICS OF ADOPTION

The divide, seen from the data side

Week 2 treated exclusion as a question about who can use a system. There is a second consequence, and it runs the other way.

Excluded means unrecorded

People the systems do not reach generate no telemetry, complete no digital courses, answer no in-app surveys. They are absent from the evidence base by construction.

Unrecorded means unserved

Investment follows evidence. A group that generates no data generates no findings, so no business case is ever written for it, so it stays excluded.

Breaking the loop

Measure the gap deliberately — coverage as a metric in its own right — and gather evidence about the excluded by means that do not require the system they cannot use.

One tile worth putting on the dashboard

Coverage: what share of the workforce this system actually reaches, reported alongside every adoption or engagement figure. It is the honest denominator, and it names the people the rest of the dashboard cannot see.

L3 · ETHICS AND CULTURE

Imported tools carry imported assumptions

82%

Language

82% of GreenLeaf employees prefer a language other than English. Hindi alone is 35%.

48%

Device

48% are Smartphone only and about 8% have only a feature phone. A desktop portal misses them.

Self-service culture

Many tools assume an employee who has desk time, an email address and the confidence to act alone.

THE TEST

Before adopting any tool, list what it assumes about the user. Then check each assumption against Employee_Master.

Three things to carry out of this room

1

Ask who is missing before you ask what the number says.

Selection, not sampling error, is what breaks HR data. If the absent people are absent for a reason connected to your question, the file cannot answer it.

2

A metric is an agreement; a dashboard is a set of incentives.

Six questions make a definition, and choosing what to display is choosing what the organization will optimize — including in ways nobody intended.

3

Analytics and surveillance are one capability.

Consent cannot carry the ethical weight in an employment relationship. Narrow collection, state purpose, limit retention, report coverage.

READING AND BRIDGE

Where to read this, and what follows

In the readings folder

Angrave et al., *HR and analytics: why HR is set to fail the big data challenge* (Unit-3 Core) · Fitz-enz & Mattox on predictive analytics · the DPDP Act 2023 · your Unit teaching note on People Analytics.

Not filed — this session is your source

Goodhart's law and the measurement-gaming literature · the workplace-surveillance debate. Kellogg, Valentine & Christin on algorithmic control (Unit-3 Core) is the closest filed text and is worth reading now rather than later.

Practice

Take one number you quoted in your Round 1 rationale. State its population, its denominator, and who is missing from it. Then say whether the claim still stands.

Coming next

Unit II — the architecture: HRIS as system of record, the ATS, the LMS, and the integration layer that decides whether any of them speak.

The fairness audit of predictive models comes in Unit III, and everything on the surveillance ladder returns there with a real model to examine.